Title

Smart Study Planner using Machine Learning Concepts

#Abstract

Time management is a common challenge faced by students. This project presents a Smart Study Planner that helps students allocate their study time efficiently. The system takes subjects, their difficulty levels, priorities, and total available study hours as input and generates an optimized study plan. A weighted decision-making approach is used to distribute time proportionally. The project also provides a graphical representation of the study plan for better understanding.

Problem Statement

Students often struggle to manage their study time effectively across multiple subjects. They may spend too much time on easier topics while neglecting difficult or important ones, leading to inefficient learning.

#Proposed Solution

The Smart Study Planner provides an automated solution that generates a balanced and personalized study schedule. It considers both the difficulty level and priority of each subject to ensure effective time distribution.

Methodology

- User inputs subjects, difficulty levels, priorities, and total study hours
- Difficulty levels are converted into numerical values:
    - Easy = 1
    - Medium = 2
    - Hard = 3
- Priority is taken on a scale of 1 to 5

- A weighted scoring mechanism is applied:
    - **Weight = Difficulty × Priority**
- Study time is allocated proportionally based on calculated weights
- The final plan is displayed along with a graphical representation

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 # Tools and Technologies Used

- Python
- Tkinter (for GUI development)
- Matplotlib (for data visualization)

 # Results

The system successfully generates a structured study plan based on user input. It ensures that subjects with higher difficulty and priority receive more study time. The graphical output enhances user understanding and usability.

 # Challenges Faced

- Handling user input errors
- Designing a fair and balanced allocation system
- Integrating visualization with GUI

 # Future Scope

- Integration with real machine learning algorithms
- Tracking student performance and adapting study plans
- Developing a mobile or web-based version
- Adding reminders and notifications

Conclusion

The Smart Study Planner is an effective tool that demonstrates how basic machine learning concepts can be applied to solve real-world problems. It helps students manage their time efficiently and improve productivity through intelligent planning.